

Research

A fused power spectrum based feature selection to identify schizophrenia from EEG signals using deep learning models: an experimental study

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Abstract

The brain, a complex and important organ in the human body, is crucial for all our body processes. For the diagnosis and ongoing monitoring of a wide spectrum of brain disorders, accurate and early detection of the proper disorder from neurophysiological monitoring methods is essential. The importance of identification of disorders like Schizophrenia in clinical practice is examined in this research, along with the difficulties in attaining accurate results, particularly when working with small structures and precise details. A novel pre-processing methodology in this stream has been implemented for further feature and knowledge extraction and subsequent image generation. With their ability to automatically extract pertinent features from input images, CNN has made a significant advancement in the domain of image classification. This study presents and investigates in details the effect of our pre-processing on various well-known CNN based architectures. Various models like DenseNet, ResNet, MobileNet, NasNet, EfficientNet and ConvNext families along with Xception, InceptionV3 and InceptionResNetV2 models have been taken into consideration. These models have become optimal approaches to various classification tasks, each providing certain benefits and addressing particular difficulties. We have conducted this research on EEG data from a standard dataset, namely, IBIB PAN - Department of Methods of Brain Imaging and Functional Research of Nervous System dataset. This study presents a thorough review of the performance of different CNN based models and their variants on our preprocessed and generated images. On comparison with state-of-the-art results we have observed that using this approach, almost all our models have exceeded the same. Medical professionals and researchers can use the outcomes of these techniques for better diagnosis and treatment planning in the field of brain disorders. Our codes will be made available at: [\[Link\]](#)

Keywords Electroencephalography (EEG) · Schizophrenia · Image Processing · Deep Learning · Power Spectrogram · Computer Vision · CNN

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1 Introduction

The human brain, an intricate nexus of neurons and synapses, holds the key to understanding the complexities of cognition and behaviour [1]. Unraveling the mysteries within this organ is an ongoing challenge, particularly in the context of diagnosing and monitoring brain disorders. This research embarks on a journey to enhance the diagnostic precision of neurophysiological monitoring methods, focusing on the classification of Schizophrenia from Brain Electroencephalogram (EEG) Signals [2]. The subtleties of this disorder, nestled within the microcosm of small neural structures, underscore the necessity for innovative approaches. In response, our study introduces a novel preprocessing methodology applied to Brain Electroencephalogram (EEG) Signals, leveraging Power Spectrogram analysis to dissect Alpha, Beta, Gamma, Delta, and Theta signals.

Schizophrenia, a complex and debilitating psychiatric disorder, remains an enigma within the realms of mental health. Characterized by disruptions in thought processes, perceptions, and emotions, it challenges both clinicians and researchers in its diagnosis and understanding. The disorder typically manifests in early adulthood, affecting individuals' cognition, emotions, and social functioning. Neurobiological factors, such as alterations in neurotransmitter systems and structural brain abnormalities, contribute to the multifaceted nature of schizophrenia. The intricacies of this disorder are reflected in its diverse symptomatology, including hallucinations, delusions, disorganized thinking, and impaired social functioning. As we delve into the intricacies of diagnosing schizophrenia, it becomes evident that the subtleties of its presentation within small neural structures necessitate innovative approaches in research and clinical practice.

Neurophysiological monitoring methods play a pivotal role in elucidating the dynamic activities of the brain, facilitating early detection and ongoing monitoring of various disorders. Schizophrenia, with its intricate details and elusive manifestations, poses a distinct challenge in achieving accurate results. Recognizing the complexities inherent in deciphering small-scale neural anomalies, our research seeks to overcome these hurdles through advanced preprocessing techniques.

Our preprocessing approach involves the fusion of Power Spectrograms into a unified image, creating a holistic representation of the diverse brain signals. This preprocessing step serves as a cornerstone for subsequent feature and knowledge extraction, augmenting the discriminative features for classification tasks. The integration of Convolutional Neural Networks (CNNs) into this methodology signifies a paradigm shift in the application of deep learning to unravel complex patterns within EEG signals. Renowned for their efficacy in image-related tasks, CNNs seamlessly align with the image-centric nature of our preprocessed EEG representations.

In this study, a meticulous exploration unfolds, delving into the nuanced impact of our preprocessing approach on a spectrum of well-established CNN architectures. These architectures include DenseNet, ResNet, MobileNet, NasNet, EfficientNet, ConvNext families, as well as individual models like Xception, InceptionV3, and InceptionResNetV2. Each model, a beacon in its own right, brings unique strengths to the table, addressing specific challenges encountered in the classification of brain disorders.

The dataset under scrutiny is sourced from the IBIB PAN - Department of Methods of Brain Imaging and Functional Research of Nervous System dataset, renowned for its comprehensive and standardized EEG data. Our research endeavors to offer a thorough and comprehensive evaluation of the performance of different CNN-based models and their variants when applied to our preprocessed and generated images.

As we navigate the intricate landscape of brain disorders, this study not only contributes to the scientific understanding of Schizophrenia but also holds promise for practical applications. Medical professionals and researchers stand to gain valuable insights for improved diagnosis and treatment planning in the realm of brain disorders.

2 Literature review

Siuly et al.[3] worked on (EMD) in preprocessing step using empirical mode decomposition. From EMD sub bands, various statistical features were extracted and later they applied ensemble bagged tree method for classification.

In another study, Jahmunah et al. [4] used nonlinear features along with support vector machine (SVM) using the kernel as radial basis function in the feature extraction and classification steps, respectively.

Devia et al. [5] have provided an event-related field features-based schizophrenia diagnosis method via EEG signals. Extremely randomized trees (ERT) features were extracted from EEG signals in this effort, and then linear discriminant analysis was used in the classification step.

In Prabhakar et al. [6], statistical features of steady-state visual evoked potentials were extracted, and in the end, classification has been executed by the k-nearest neighbours (KNN) method.

Li et al. [7] used solitary pulmonary nodule features and SVM classification for schizophrenia diagnosis via EEG signals.

In another study, Shim et al. provided a new method of schizophrenia diagnosis via EEG signals [8]. This investigation used sensor-level and source-level features in the feature extraction step and then employed the Fisher's score for feature selection. Ultimately, the SVM method was used in the classification step, and they achieved promising results.

Using EEG data from patients in the resting state, [9] investigated how well-suited typical machine learning algorithms were to produce classifiers for schizophrenia. Their method involves dividing the EEG data into sliding windows and computing a range of linear and non-linear metrics over these windows using a processing pipeline. Nearly every machine learning algorithm they examined, including SVM (0.89), RF (0.87), LR (0.86), kNN (0.86), and DT (0.68), produced results with strong classification power.

Shoeibi et al.'s study, [10], used the CNN-LSTM model to examine EEG signals in order to diagnose schizophrenia.

Using an eXtreme Gradient Boosting (XGB) algorithm, Carmen et al.'s work [11] focuses on classifying patients with schizophrenia. Because it can improve speed and execution in the ML model, XGB, a supervised method that is a variation of gradient boosting, is used.

Two steps are involved in the research study by Jie Sun et al. [12]. Prior to being converted into a series of red, green, and blue (RGB) images that convey spatial information, the EEG time series is preprocessed in order to extract time-domain and frequency-domain properties. Second, we build hybrid deep neural networks (DNNs) to categorize schizophrenia patients and healthy controls based on RGB images by combining long short-term memory and convolution neural networks.

Conventional machine learning (ML) and deep learning (DL) techniques are used in this field of AI research [13, 14]. The method for diagnosing schizophrenia using artificial intelligence comprises preprocessing portions, feature extraction and selection, and classification in the conclusion.

The most crucial step in using EEG signals to diagnose schizophrenia is feature extraction. The extracted features from EEG signals in conventional machine learning are primarily divided into four groups: non-linear [14] fields, time-frequency [15], frequency [16], and time [17].

A dataset of functional magnetic resonance imaging (fMRI) was utilized to identify first-episode schizophrenia using an ensemble technique [18]. Using that strategy, a classification accuracy of 84.7% was achieved.

Using kNN and cortical characteristics collected from MRI data, a classification accuracy of 85% for patients with autism and schizophrenia was reached in [19].

Zhu et al. achieved an accuracy of 76% in identifying schizophrenia at early clinical stages using an SVM classifier based on numerous structural MRI features [20].

Using brain-wise functional connectivity characteristics from MRI data, SVM was also utilized to classify schizophrenia with an accuracy of 85% [21].

A deep canonically-correlated sparse autoencoder was proposed by Li et al. [22] to distinguish between patients with schizophrenia and healthy controls. Their technique was used on fMRI as well as single nucleotide polymorphisms. Using genetic data, an accuracy of 95.65% was attained, whereas fMRI yielded an accuracy of 80.53%. Schizophrenia has also been categorized using textual data.

Several linguistic variables were retrieved from social media posts in the study article on schizophrenia identification [23]. A Random Forest technique was then applied to achieve a classification accuracy of 96%.

Demographic data and clinical assessment questionnaires dataset were used in [24] and finally tested and trained using SVM classifier which achieved an accuracy of 90%.

Buettner et al. [25] achieved 71% of accuracy in classifying schizophrenia using the random forest classifier based on power features extracted from several frequency bands.

From 5-minute resting-state EEG signal, twelve features were extracted from five EEG bands and finally were used in a SVM model allowing Tikka et al. to achieve an accuracy of 79% in classifying schizophrenia patients from healthy controls [26].

A research article [27] identified the EEG bands that are associated with bipolar disorder using Random Forest and Ant Colony Optimization technique.

Siyuli et al. [28] in their work introduces a deep residual network (deep ResNet) based feature extraction design that can automatically extract representative features from EEG signal data for identifying schizophrenia.

In other domains, Kasim et al. [29] proposed a methodology using Multitaper and Multivariate Variational Mode Decomposition, for feature extraction and then using deep learning models for the detection of Attention Deficit Hyperactivity Disorder (ADHD).

Similarly, Lee et al. [30], proposed a ResNet-LSTM hybrid model for predicting epileptic seizures using a pretrained model with supervised contrastive learning. Also an EEG signal-based classification of mental tasks using a one-dimensional ConvResT model was proposed by Manasa et al. [31].

Kasim & Tosun et al. [32] proposed an effective removal of eye-blink artifacts in EEG signals with semantic segmentation system. Also the work of Bera et al. [33] gives a comprehensive overview on the medical applications of electroencephalography (EEG).

3 Proposed methodology

3.1 Dataset description

For this work related to brain signal analysis for identifying Schizophrenic patients, we have taken into consideration the publicly available dataset, IBIB PAN - Department of Methods of Brain Imaging and Functional Research of Nervous System dataset available at <https://tinyurl.com/ymxn3zvb>. The dataset consists of 28 EEG recordings (14 healthy and 14 schizophrenia patients). These signals were preprocessed using bandpass filtering for Alpha, Beta, Gamma, Delta, and Theta frequency bands. We have split the datasets into training, validation and test sets (70% as the training data, 10% as validation data and 20% as test data). our dataset was split subject-wise, meaning that EEG recordings from a single individual were not shared between training and test sets.

We extracted the raw data from the edf files available using the mne python module. This raw data was then clipped for the Alpha, Beta, Gamma, Delta and Theta brain signal ranges respectively. The ranges for each of these signals are provided in Table 1.

Fast Fourier Transform (FFT) was then applied to the resultant bandpass data available after clipping. The resultant data were splitted into intervals of 5000, 10,000, 15000 and so on. We then used the Librosa module to generate the Power Spectrogram of each of these splits. The spectrograms of each signals were then taken together and combined into five different sections of a single image using the Matplotlib Python library. Two sample of these images are as shown in Fig. 1 and Fig. 2

The original dataset consisted of 28 edf files containing the EEG signals of 28 different patients as shown in table Table 2.

From these data a total of 2487 such images were generated and collected of various intervals. They are then kept in their respective directories the hierarchy of which is represented in Fig. 3. The distribution of images are as given in Table 3. The hierarchy depicted in Fig. 3 represents the final structure after ensuring this separation. The number represents the numerical count of the images and not the subject number.

Table 1 Bandpass filter ranges of each brain signal

Brain Signals	Bandpass Filter Range
Delta	(0.5, 4)
Theta	(4, 8)
Alpha	(8, 12)
Beta	(13, 30)
Gamma	(30, 80)

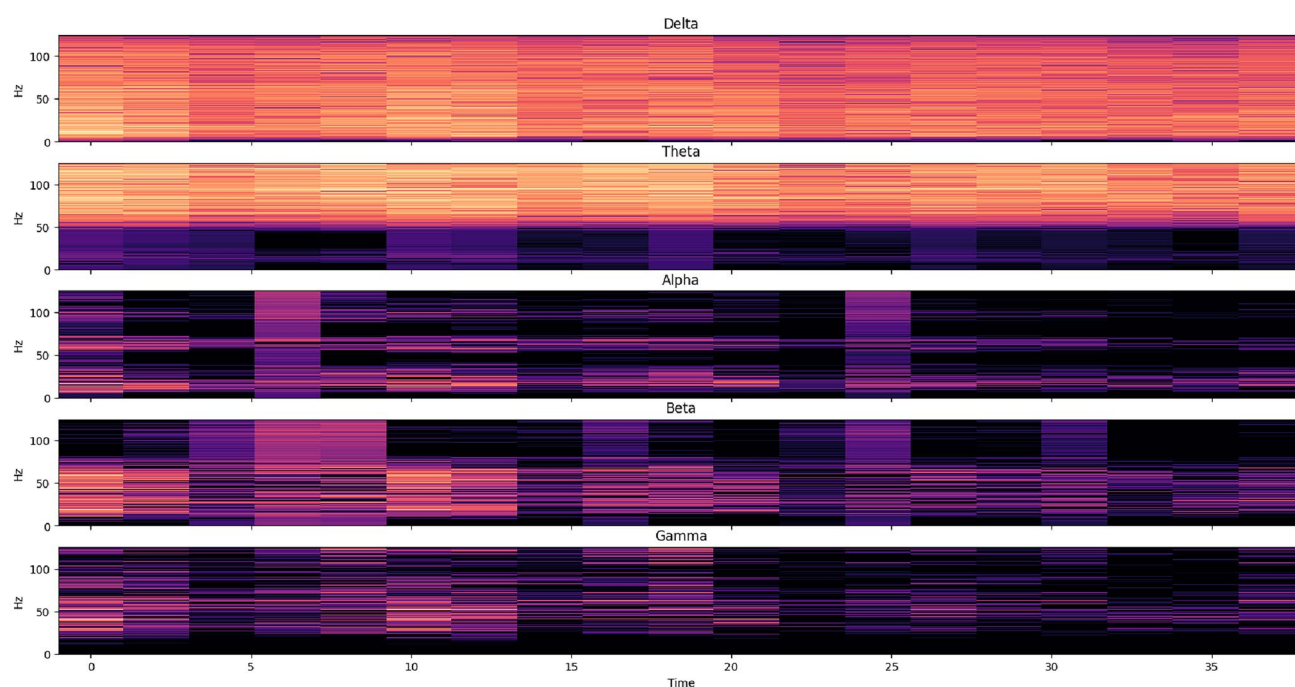


Fig. 1 The Power Spectrograms of a Healthy Patient, highlighting the Alpha, Beta, Gamma, Delta and Theta Signals. The textual labels and axes in the figures are visualization aids and are not part of the actual input to the deep learning models. The processed spectrograms used for training are free from such artifacts

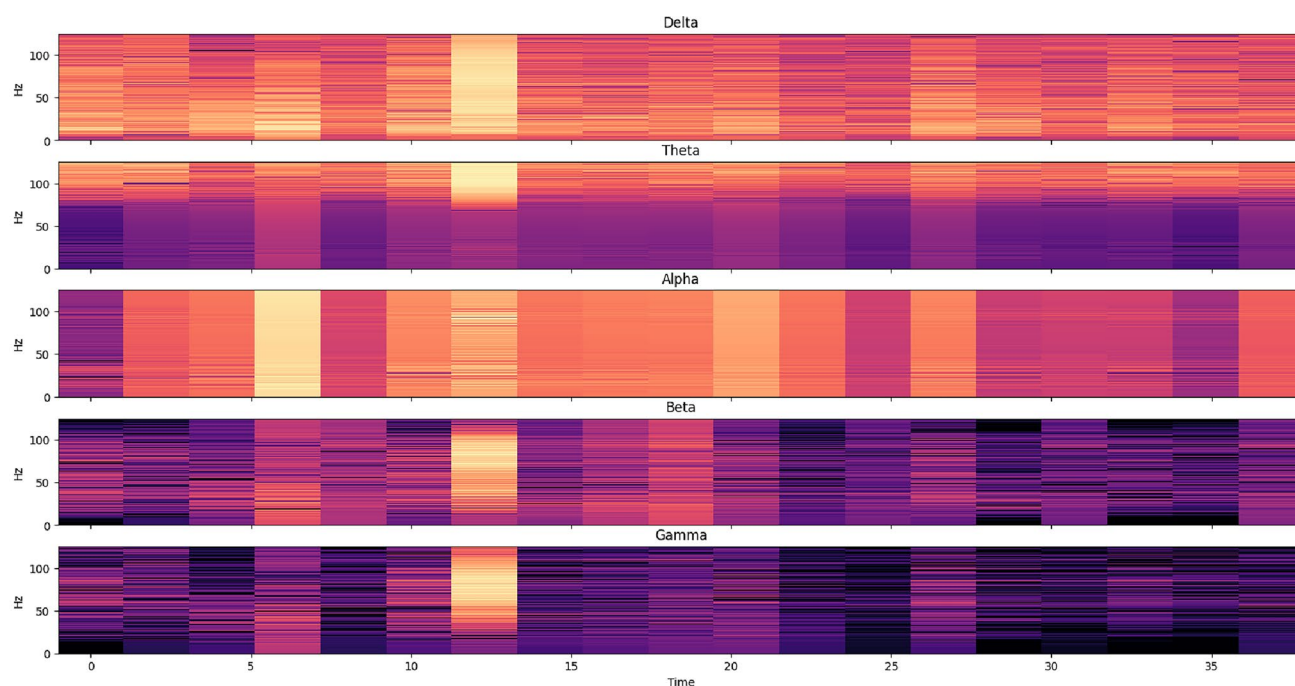
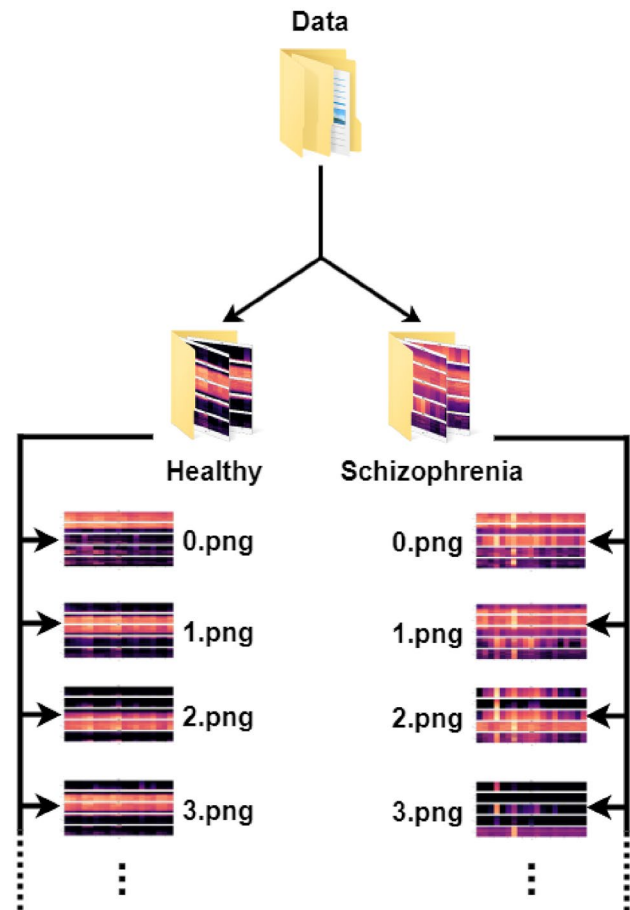


Fig. 2 The Power Spectrograms of a Schizophrenic Patient, highlighting the Alpha, Beta, Gamma, Delta and Theta Signals. The textual labels and axes in the figures are visualization aids and are not part of the actual input to the deep learning models. The processed spectrograms used for training are free from such artifacts

Table 2 Distribution of edf files in the original dataset

Dataset	Classes	No. of edf files
IBIB PAN Dataset	Healthy	14
	Schizophrenic	14

Fig. 3 The hierarchy of our final dataset**Table 3** Distribution of images in our final dataset

Dataset	Classes	No. of images
IBIB PAN Dataset	Healthy	1070
	Schizophrenic	1417

3.2 Feature extraction

Feature extraction is a necessity for signal processing. Here brain signal features are expressed as mathematical algorithms, which in turn are used to extract important information which is not observable in the raw data. The features are extracted from the spectral or frequency domain and temporal or time domain. A spectrogram is generally used for this work as it contains both spectral and temporal information.

3.2.1 Fast Fourier transform (FFT)

The Fast Fourier Transform (FFT) is a widely utilized algorithm in signal processing, playing a pivotal role in transforming time-domain signals into their frequency-domain representations. Its origins trace back to the Cooley-Tukey algorithm,

developed in the 1960s [34], which significantly optimized the computation of discrete Fourier transforms; and to Joseph Fourier's work in 1822 [35]. In this research, we employed Fast Fourier Transform (FFT) on electroencephalogram (EEG) data stored in European Data Format (EDF) files to analyze the frequency components of brain signals. The FFT algorithm was applied to convert the EEG time-domain signals into the frequency domain, revealing the distribution of power across different frequency bands. The mathematical formulation of the FFT, defined in Eq. 1

$$X(f) = \int_{-\infty}^{\infty} x(t) * e^{-j2\pi ft} dt \quad (1)$$

allowed us to efficiently decompose the complex EEG signals into their constituent sinusoidal components. This spectral analysis facilitated the identification of characteristic frequency patterns associated with cognitive processes, providing valuable insights into the neurophysiological mechanisms underlying various mental states. Our utilization of FFT on EDF files contributes to the burgeoning field of EEG signal processing, enhancing our understanding of brain function and paving the way for advancements in neuroscientific research and clinical applications.

3.2.2 Spectrogram

A spectrogram is a visual representation of the frequency spectrum of a signal as it changes with time. For audio signals, it's also called sonographs, voiceprints or voicegraphs. For speech signals the spectral analysis is conducted by using the Fast Fourier Transform algorithm to identify the Discrete Fourier Transform (DFT) of the samples in the frames.

3.2.3 Power spectrogram

In this study, we harnessed the Power Spectrogram analysis technique to delve into the temporal and spectral characteristics of EEG data on each of the five brain signals, extracted from European Data Format (EDF) files. The Power Spectrogram, an evolution of the Fourier Transform, provides a detailed visualization of the distribution of signal power across both time and frequency domains. Originating from the Short-Time Fourier Transform (STFT), the Power Spectrogram overcomes the limitation of assuming stationarity in the signal by employing a window function to capture the signal's dynamics over short time intervals. The formula for the Power Spectrogram is given in

$$P(f, t) = |STFT[x(t)]|^2 \quad (2)$$

where $x(t)$ is the input $STFT[x(t)]$ represents the Short – Time Fourier Transform of the signal

By calculating the squared magnitude of the STFT, the Power Spectrogram provides a robust representation of signal power at different frequencies over time, offering a nuanced perspective on the dynamic nature of brain activity. This methodology enriches our understanding of EEG data, enabling the identification of transient changes in spectral content associated with cognitive processes or neurological disorders.

3.3 Reasons for fusing the spectrograms together

In our research, the innovative strategy of fusing power spectrograms from distinct frequency bands, namely delta, theta, alpha, beta, and gamma, within different partitions of a single image was adopted to capture comprehensive information about the diverse frequency components of EEG signals. This approach capitalizes on the spatial context inherent in images, leveraging the intrinsic relationships between different brainwave frequencies.

Each partition of the image encapsulates the nuanced power distribution of a specific frequency band, facilitating the extraction of multi-scale features. By feeding these multi-part images into Convolutional Neural Network (CNN)-based models, we aimed to exploit the spatial hierarchies embedded in the amalgamated spectrogram representation. This method not only optimally utilizes the spatial correlations among distinct frequency components but also allows for the model to discern intricate patterns that may be crucial for understanding complex cognitive processes or pathology.

The fusion of these spectrograms into a unified image structure represents a novel and effective approach, enhancing the discriminative power of our CNN models in the analysis of EEG data and contributing to the advancement of

In our work, after the above preprocessing that we have conducted, the images were then normalized by dividing them with 255 followed by attaching weights to the classes, so as to remove the class imbalancing. The results obtained by the models after the proposed preprocessing has been provided in details in section 5

In this section, we describe the architecture of all the models that we have considered for the classification of Schizophrenia patients from EEG signals.

A convolutional neural network architecture called Xception was proposed by Chollet et al. [36] in 2017. It is an expansion of the Inception architecture and is known as “Extreme Inception” since it substitutes depthwise separable convolutions for the regular Inception convolutions. The core of the Xception architecture consists of depth-wise separable convolution layer blocks, followed by batch normalization and ReLU activation. A pointwise convolution layer that implements a 1x1 convolutional filter for combining the output channels of the depth-wise convolution and a depthwise convolution layer that employs a single convolutional filter to each input channel make up the two separate layers which make up the depthwise separable convolution layers. The network’s final layer consists of a fully connected layer for classification and a layer for global average pooling. The residual connections allow for richer network topologies while addressing the vanishing gradient problem, while the depthwise separable convolutions reduce overfitting and are more computationally efficient than standard convolutions. As a result, Xception is a strong and successful model for tasks involving image classification. Figure 4 illustrates the architecture of the Xception model and Fig. 5 depicts how it has been used in our model.

ResNet152 stands as a pinnacle in the realm of deep convolutional neural networks, particularly designed for image classification endeavors. Originating from the research laboratories of Microsoft Research Asia, it is an extension of the ResNet architecture and was introduced by Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun in their landmark 2016 paper, "Deep Residual Learning for Image Recognition" [37]. This architecture builds upon the success of its predecessors, ResNet50 and ResNet101, by further deepening the network to an impressive 152 layers. Retaining the ingenious residual connections, ResNet152 continues to address challenges associated with training deep networks [38], such as the vanishing gradient problem. Its heightened depth enhances feature extraction and representation capabilities, making it an invaluable tool in diverse applications beyond image classification, spanning fields like object detection, segmentation, and transfer learning. ResNet152 has proven to be an influential model, showcasing exceptional performance in various benchmarks and competitions, solidifying its status as a cutting-edge solution in advancing the landscape

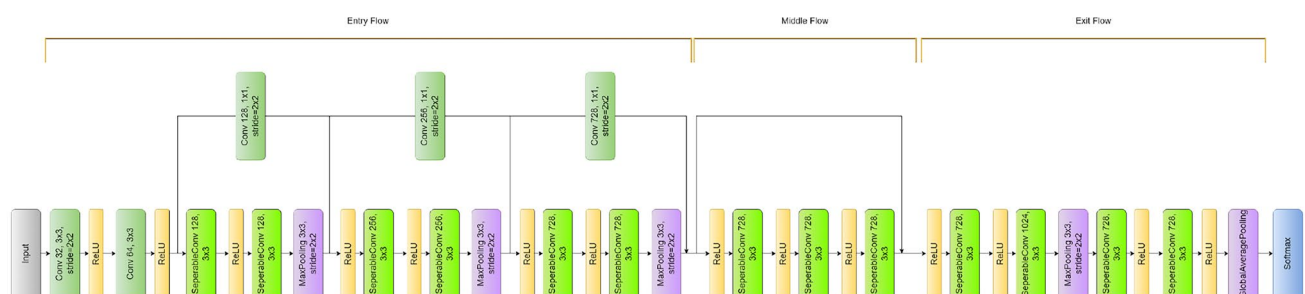


Fig. 5 Architecture of our model along with Xception model

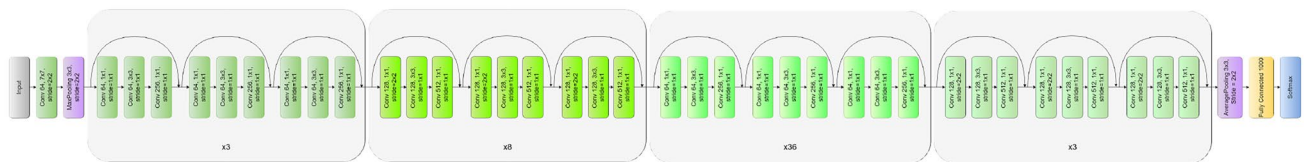
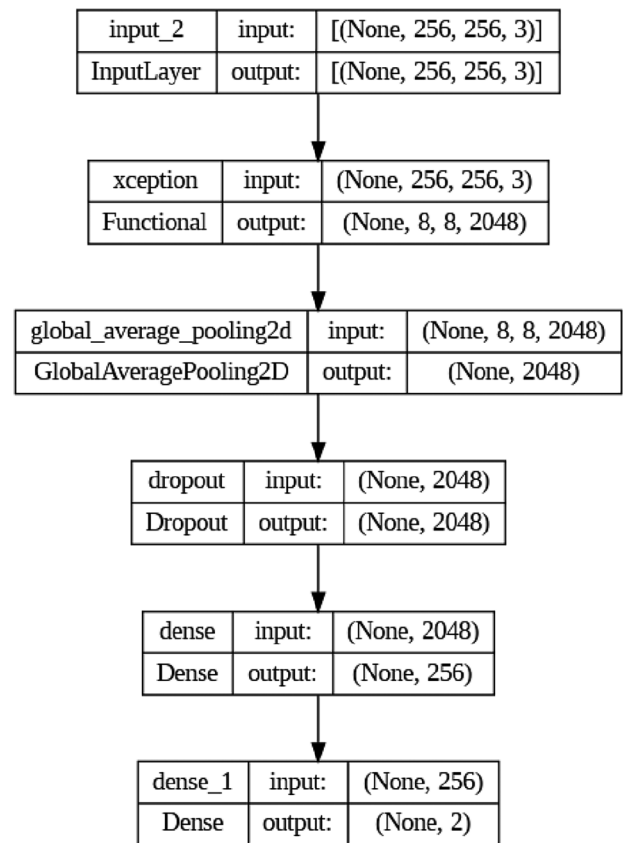


Fig. 6 Architecture of the ResNet152 model

of computer vision. The picture in Fig. 6 shows the architecture of ResNet152 and Fig. 7 shows the architecture of our model using ResNet152.

4.3 InceptionV3

InceptionV3, a pivotal convolutional neural network (CNN) architecture for image classification, emerged from the research laboratories of Google. Developed by Christian Szegedy, Vincent Vanhoucke, Wei Liu and others, InceptionV3 is part of the Inception family of models. Introduced in the landmark 2015 paper, "Going Deeper with Convolutions" [39], this model is recognized for its innovative inception modules, featuring diverse kernel sizes within the same layer to capture multi-scale features effectively. InceptionV3 strikes a balance between computational efficiency and model accuracy, making it suitable for a wide range of applications beyond image classification, including object detection and segmentation. Its versatility has led to widespread adoption in computer vision research, solidifying InceptionV3 as a significant contributor to the ongoing advancements in deep learning methodologies. Figure 8 displays the architecture of InceptionV3 and Fig. 9 displays the architecture of our model using InceptionV3.

Fig. 7 Architecture of our model along with ResNet152 model

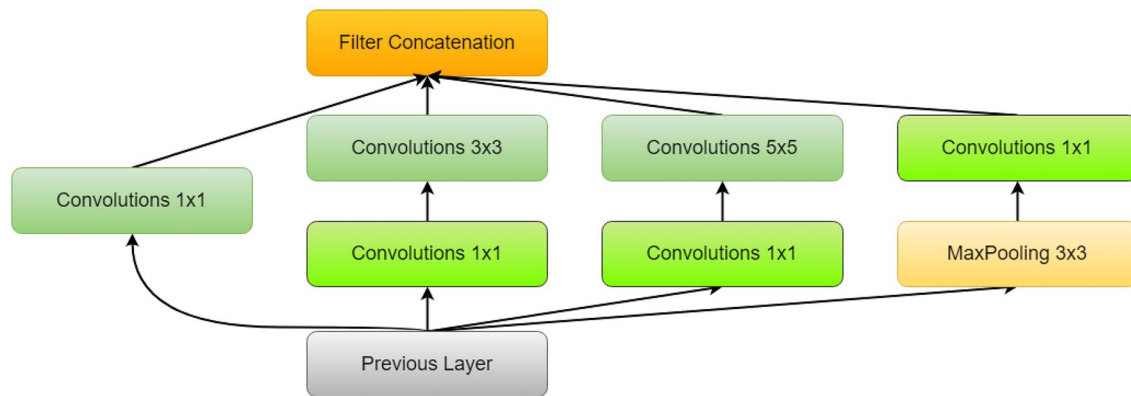
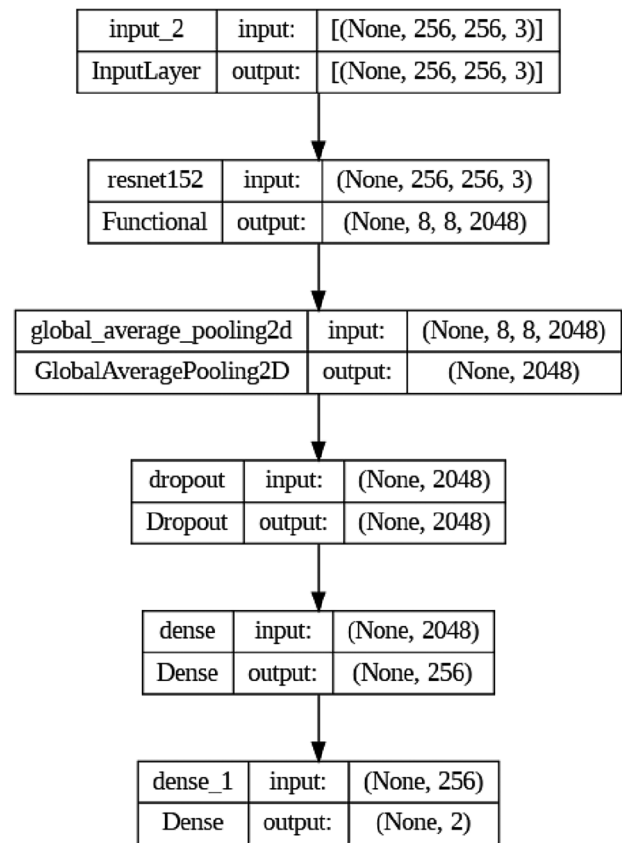


Fig. 8 Architecture of the InceptionV3 model

4.4 InceptionResNetV2

InceptionResNetV2, which was introduced by Szegedy et al. in 2017 [40], combines the advantages of the Inception and ResNet architectures. It makes use of residual connections with Inception modules, which solve the vanishing gradient issue as well as allow for more intricate network designs. The model can quickly learn and extract characteristics from incoming data and needs fewer parameters to attain high accuracy. The InceptionResNetV2 architecture is illustrated in Fig. 10 and Fig. 11 depicts its usage in our model.

Fig. 9 Architecture of our model along with InceptionV3 model

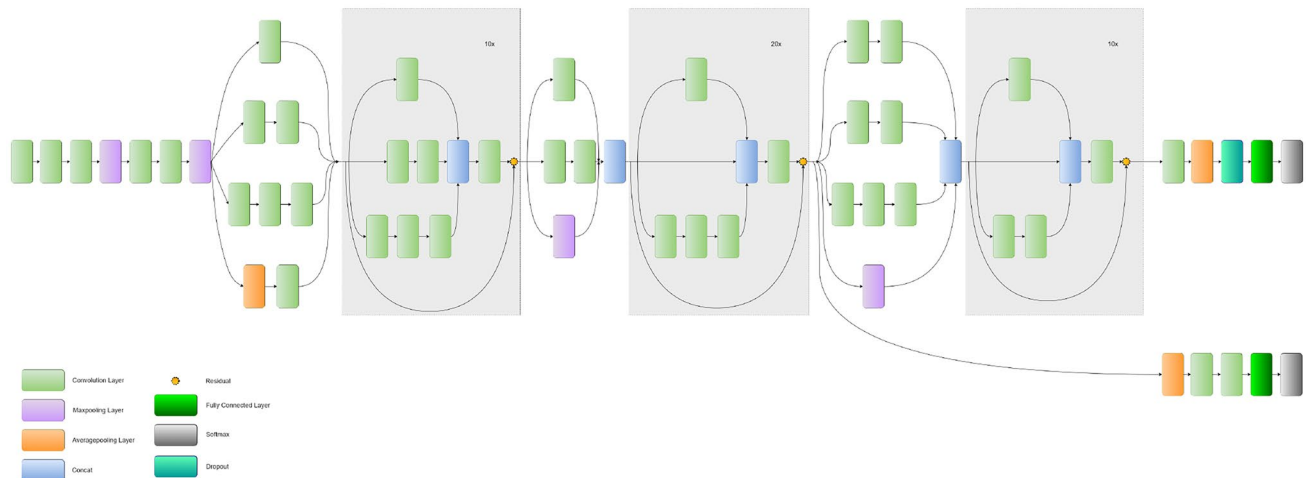
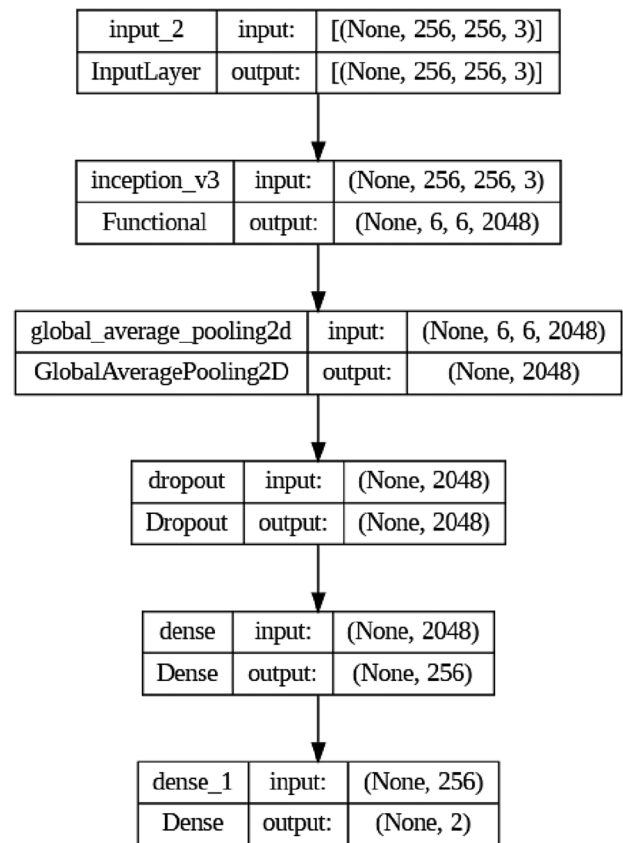


Fig. 10 Architecture of the InceptionResNetV2 model

4.5 MobileNetV2

Convolutional Neural Network MobileNetV2, developed by Sandler et al. in 2018 [41], is intended for mobile and embedded vision applications. In order to increase efficiency while maintaining representational power, it makes use of depthwise separable convolution, inverted residuals, and linear bottlenecks. It consists of two different kinds of blocks: downsizing blocks with a stride of 2, and blocks with a stride of 1 for residual blocks. The blocks of both types have three levels. Each block's initial layer is an 1x1 convolution with ReLU6 activation. A depthwise convolution is

Fig. 11 Architecture of our model along with Inception-ResNetV2 model

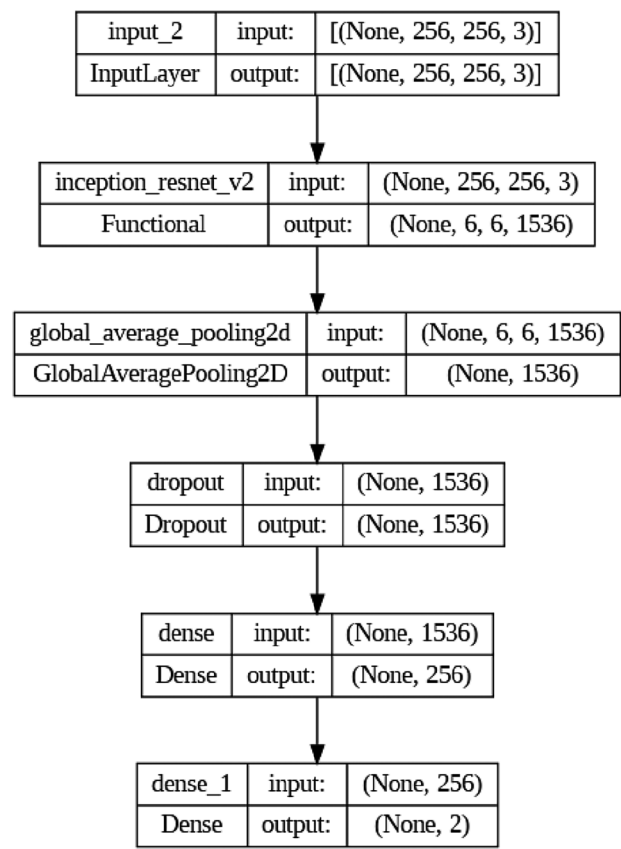
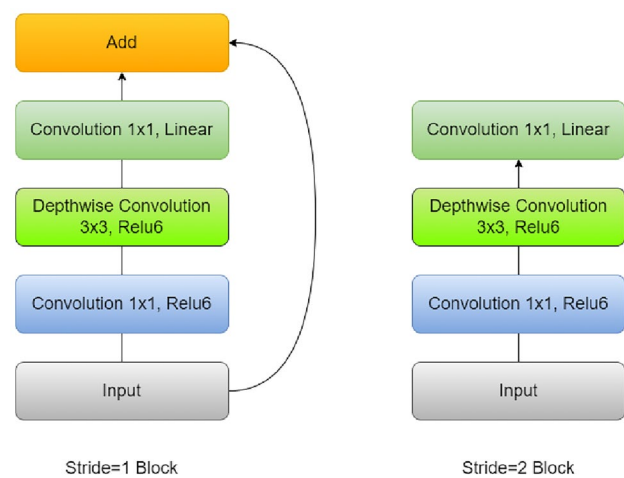
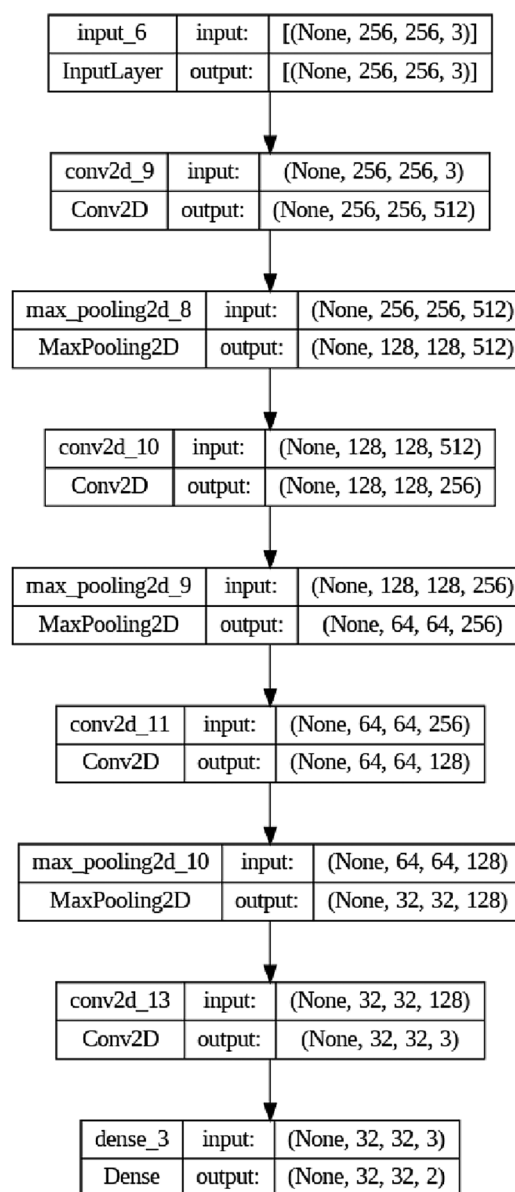


Fig. 12 Architecture of the MobileNetV2 model



the second layer, and it applies a different convolutional filter to each input channel. Each block's last layer is an 1x1 convolution once more, but this time there is no non-linearity. In this network, a width multiplier is utilized to adjust the network for various hardware and resource limitations. Due to its great efficiency and minimal weight, MobileNetV2 is an ideal model for use in contexts with limited resources, such as mobile devices. MobileNetV2 performs well on a variety of vision tasks, including semantic segmentation, object detection, and image classification, despite its modest size. The architecture of MobileNet is visualised in Figs. 12, 13 depicts its usage in our model.

Fig. 13 Architecture of our model along with Mobile-NetV2 model



4.6 DenseNet201

DenseNet201 is a well-liked deep learning architecture that is frequently used for image classification problems [42]. It was first introduced by Gao Huang et al. [43]. Because DenseNet can effectively use feature maps and lessen the vanishing gradient issue that frequently arises in deep neural networks, it is recommended over alternative architectures. In order to accomplish this, skip connections are added between each layer, enabling the network to access feature maps from all earlier layers. Better precision, quicker convergence, and a more effective utilization of the parameters are the outcomes of this. Besides achieving state-of-the-art results in various image classification tasks [44] [45], DenseNet has also demonstrated effectiveness in handling limited training data [46][47]. Figure 14 visualises the architecture of DenseNet201 and Fig. 15 depicts its usage in our model.

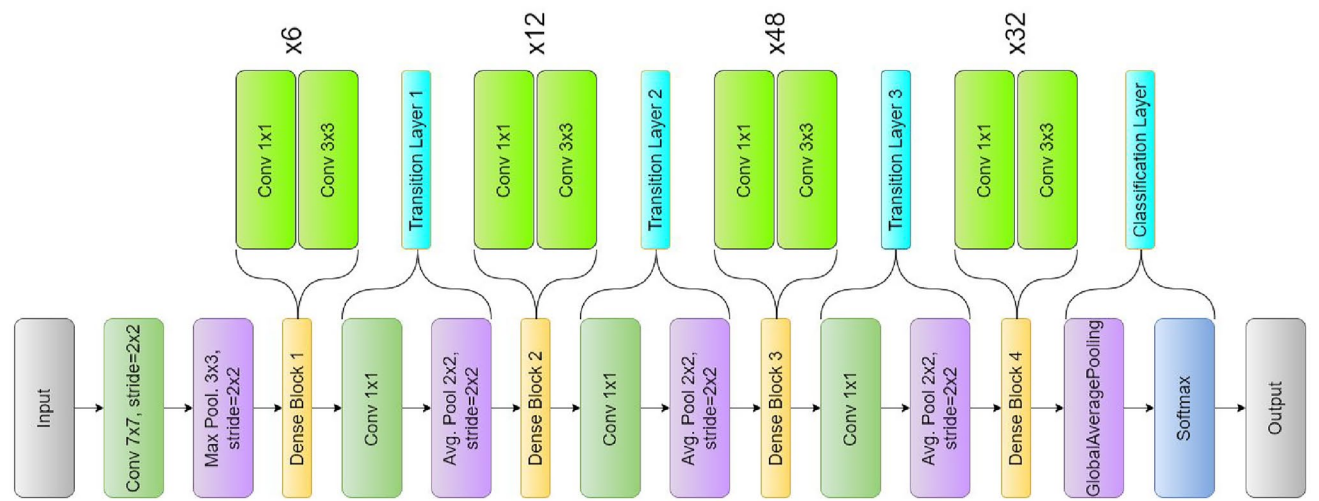
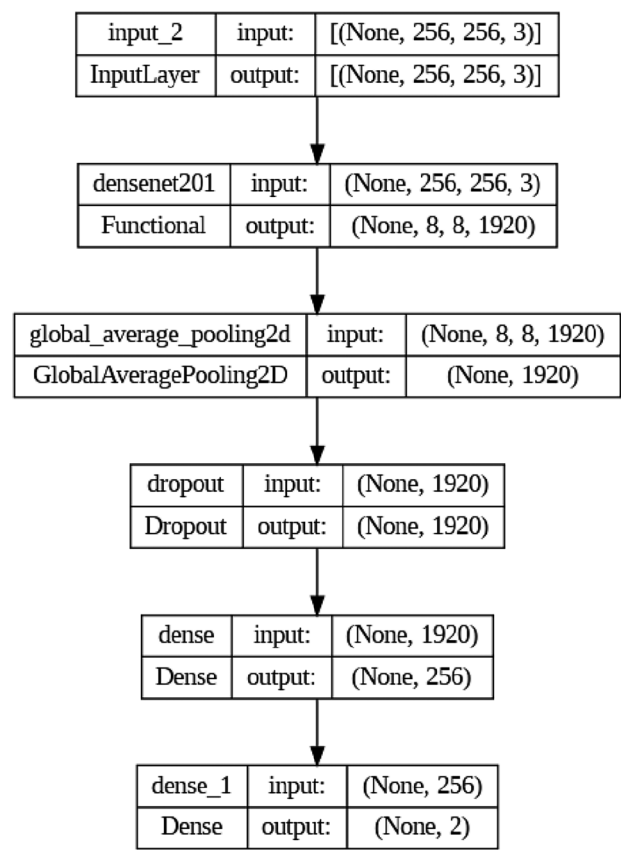


Fig. 14 Architecture of the DenseNet201 model

Fig. 15 Architecture of our model along with DenseNet201 model



4.7 NASNetLarge

NasNetLarge, a pioneering convolutional neural network architecture for image classification [48, 49], emerged from the field of neural architecture search (NAS) and was developed by Google’s AI research team. Introduced in the groundbreaking 2018 paper, "Learning Transferable Architectures for Scalable Image Recognition", [50] authored by Barret Zoph, Vijay Vasudevan, Jonathon Shlens, and Quoc V. Le, NasNetLarge stands out for its unique approach to architecture design. Leveraging reinforcement learning and evolutionary algorithms, NasNetLarge autonomously

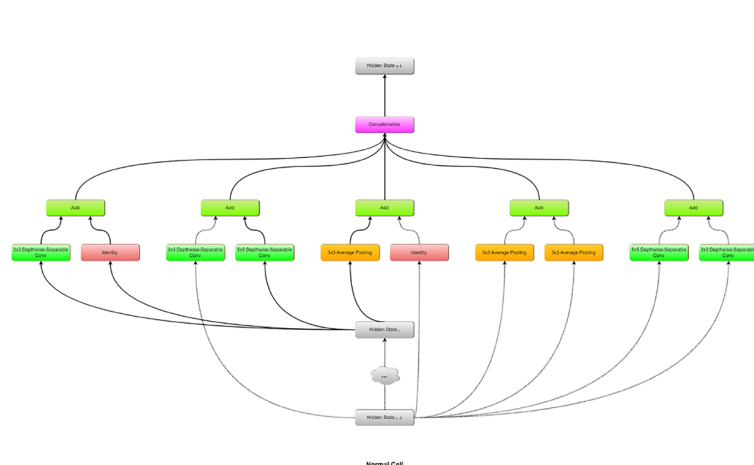
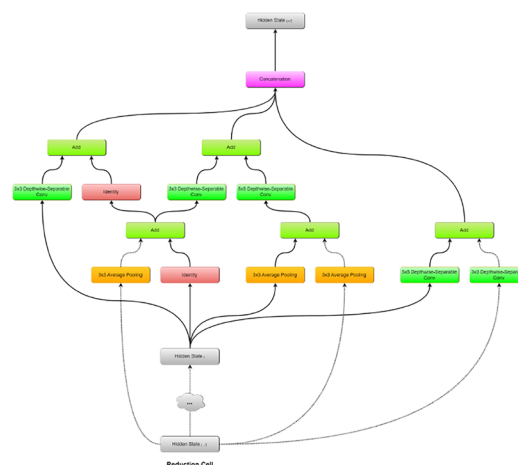


Fig. 16 Architecture of the NASNetLarge model

Fig. 17 Architecture of our model along with NASNet-Large model



input_2	input:	[(None, 256, 256, 3)]
InputLayer	output:	[(None, 256, 256, 3)]

NASNet	input:	(None, 256, 256, 3)
Functional	output:	(None, 8, 8, 4032)

global_average_pooling2d	input:	(None, 8, 8, 4032)
GlobalAveragePooling2D	output:	(None, 4032)

dropout	input:	(None, 4032)
Dropout	output:	(None, 4032)

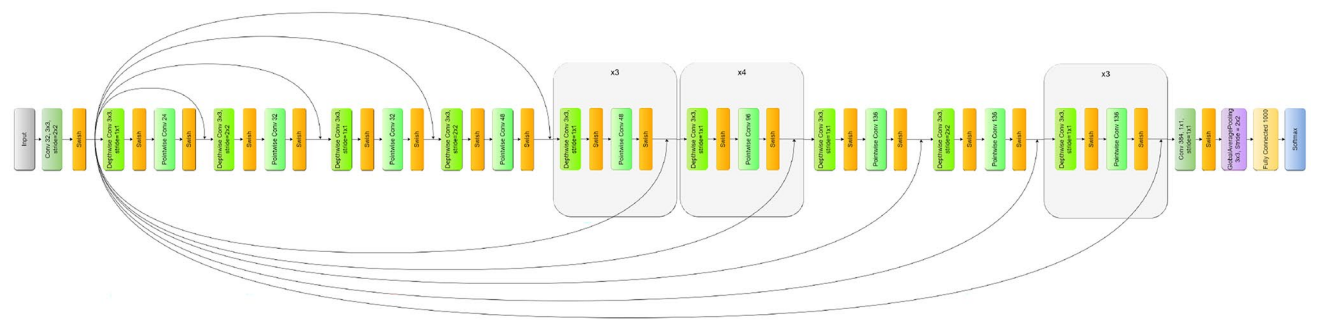
dense	input:	(None, 4032)
Dense	output:	(None, 256)

dense_1	input:	(None, 256)
Dense	output:	(None, 2)

identifies optimal network architectures, resulting in models that outperform many hand-crafted designs [51]. Its adaptability extends beyond image classification, making it a promising solution for various computer vision tasks and serving as a testament to the potential of automated architecture search in advancing deep learning methodologies [52, 53]. Figures 16, 17 depicts the architecture of the NASNetLarge model and its usage in our model respectively.

4.8 EfficientNetB4

EfficientNetB4, a noteworthy convolutional neural network architecture for image classification, stems from the concept of efficient model scaling and was introduced by Mingxing Tan and Quoc V. Le in their 2019 paper, "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks" [54]. This model belongs to the EfficientNet family, which emphasizes optimal trade-offs between model size and performance. EfficientNetB4 is characterized by its



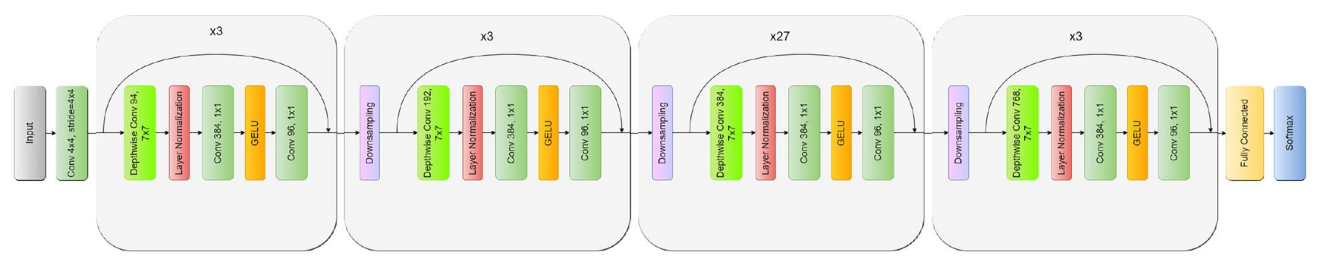
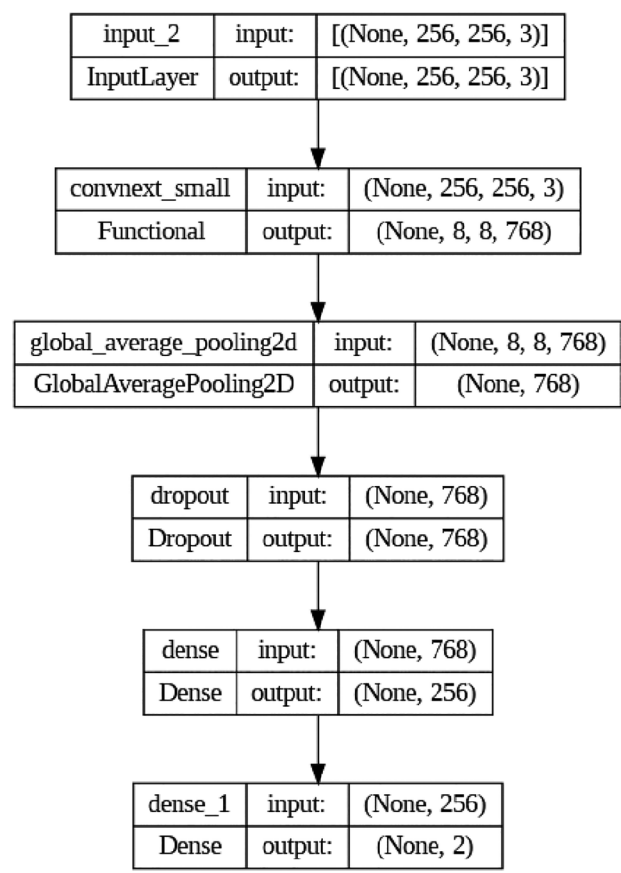


Fig. 20 Architecture of the ConvNeXtSmall model

Fig. 21 Architecture of our model along with ConvNeXtSmall model



optimally reuses feature maps, reducing computational complexity. This architectural innovation not only proves its worth in image classification but also finds applicability in various computer vision fields [61, 62]. ConvNeXtSmall's emphasis on computational efficiency and effective feature reuse marks it as a promising tool for researchers and practitioners seeking to strike a balance between model performance and resource utilization in deep learning applications. Figures 20, 21 a pictorial representation of the architecture of the ConvNeXtSmall model and its usage in our model respectively.

5 Results and discussion

This section presents the comprehensive findings and the analysis that goes along with the suggested pre-processing in relation to our CNN models that are utilized to detect schizophrenia using brain electroencephalogram (EEG) information. Table 3 already provides the distribution of photos in the dataset. There is also a discussion of the discovered results'

ramifications. Furthermore, we provide a comparative analysis to demonstrate that the suggested approach performs better than alternative models and widely-applied strategies employed in the literature review.

5.1 System configuration

The entire series of experiments has been conducted on a Jupyter Notebook provided by Google's collaborative environment, which is outfitted with a 12 GB NVIDIA Tesla T4 GPU. We have used Python 3 to construct our suggested strategy for the tests; Tensorflow, Keras, Matplotlib, Scikit, Numpy, Librosa and Pandas are the main open-source modules used.

5.2 Evaluation metrics

Metrics for evaluation are crucial in determining a predictive/learning model's potency and usefulness. In other words, these measures aid in quantifying the model's performance in terms of its accuracy in result prediction. To acquire a thorough view of the model's performance and make sure it satisfies the criteria of the problem under examination, it is critical to apply a number of evaluation metrics. These metrics are used to assess how well a system performs in predicting the correct class label for an input in classification challenges. Let's have a look at a situation with two classes where one class is designated as "positive" and the other as "negative."

- True Positive (TP) refers to the number of accurately categorized positive samples.
- False Positive (FP) refers to the number of negative samples that are mistakenly labeled as positive.
- False Negative (FN) refers to those samples that are mistakenly labeled as negative despite being positive.
- True Negative (TN) refers to the number of accurately categorized negative samples.

For the present work, we use the following performance metrics:

Accuracy The percentage of events that are correctly predicted over all instances is known as accuracy. It is employed to assess how well a model performs in terms of producing accurate predictions. It is determined by dividing the total number of predictions made by the model by the number of predictions that were accurate. Mathematically, Eq. 3 represents the accuracy.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

Precision The proportion of accurate positive predictions amongst all positive predictions made by a model is known as its precision. It is determined by dividing TP by the product of TP and FP. Mathematically, precision can be represented as in Eq. 4.

$$Precision = \frac{TP}{TP + FP} \quad (4)$$

Recall Recall is the percentage of actual positive events that the model correctly identifies. It is the proportion of TP to TP plus FN as shown in Eq. 5. A higher recall demonstrates that the model is better at identifying positive examples, whereas a lower recall raises the possibility that some positive cases may have been missed by the model.

Table 4 Hyperparameters of the models

Hyperparameter	Value/Name
Optimizer	Adam [63]
Loss function	Sparse Categorical Cross Entropy
Learning rate	0.001
No. of epochs	60

Table 5 Performance measure of each model along with their total number of parameters

Model	No. of parameters	Accuracy (%)	Precision (%)	Recall (%)
Xception	21,386,538	99.79	99.83	99.74
ResNet152	58,896,002	98.54	98.54	98.47
InceptionV3	22,327,842	99.38	99.39	99.34
InceptionResNetV2	54,730,722	98.96	98.80	99.10
MobileNetV2	2,586,434	99.79	99.82	99.79
DenseNet201	18,814,274	100.00	100.00	100.00
NASNetLarge	85,949,780	99.38	99.24	99.48
EfficientNetB4	18,133,345	99.58	99.65	99.49
ConvNeXtSmall	49,652,066	100.00	100.00	100.00

Table 6 Class-wise and overall results obtained by the models

Models	Classes	Pre.	Rec.	F1	Acc.	Overall Acc.	Overall Pre.	Overall Rec.	Overall F1
Xception	Healthy	1.0000	0.9948	0.9974	0.9948	0.9979	0.9983	0.9974	0.9978
	Schizophrenia	0.9965	1.0000	0.9983	1.0000				
ResNet152	Healthy	0.9850	0.9801	0.9825	0.9801	0.9854	0.9854	0.9847	0.9850
	Schizophrenia	0.9857	0.9892	0.9875	0.9892				
InceptionV3	Healthy	0.9952	0.9905	0.9928	0.9905	0.9938	0.9939	0.9934	0.9936
	Schizophrenia	0.9926	0.9963	0.9945	0.9963				
InceptionResNetV2	Healthy	0.9760	1.0000	0.9878	1.0000	0.9896	0.9880	0.9910	0.9894
	Schizophrenia	1.0000	0.9819	0.9909	0.9819				
MobileNetV2	Healthy	1.0000	0.9952	0.9976	0.9952	0.9979	0.9982	0.9976	0.9979
	Schizophrenia	0.9963	1.0000	0.9982	1.0000				
DenseNet201	Healthy	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	Schizophrenia	1.0000	1.0000	1.0000	1.0000				
NASNetLarge	Healthy	0.9848	1.0000	0.9923	1.0000	0.9938	0.9924	0.9948	0.9935
	Schizophrenia	1.0000	0.9895	0.9947	0.9895				
EfficientNetB4	Healthy	1.0000	0.9898	0.9949	0.9898	0.9958	0.9965	0.9949	0.9957
	Schizophrenia	0.9930	1.0000	0.9965	1.0000				
ConvNeXtSmall	Healthy	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	Schizophrenia	1.0000	1.0000	1.0000	1.0000				

$$Recall = \frac{TP}{TP + FN} \quad (5)$$

F1 Score The F1 score, which integrates a model's recall and precision into a single value, is a common assessment metric for imbalanced data in a classification task. It is calculated by summing the harmonic means of recall and precision as in Eq. 6. Higher value of F1 Score indicates better performance.

$$F1 = \frac{2 * Precision * Recall}{Precision + Recall} \quad (6)$$

5.3 Implementation

Table 4 shows the values of the different hyperparameters that we used for this study. The transfer learning models architectures have already been trained on the large-scale ImageNet dataset. Therefore, the learned features from ImageNet help to capture the inherent feature information from the images in the dataset. The Adam optimizer has been

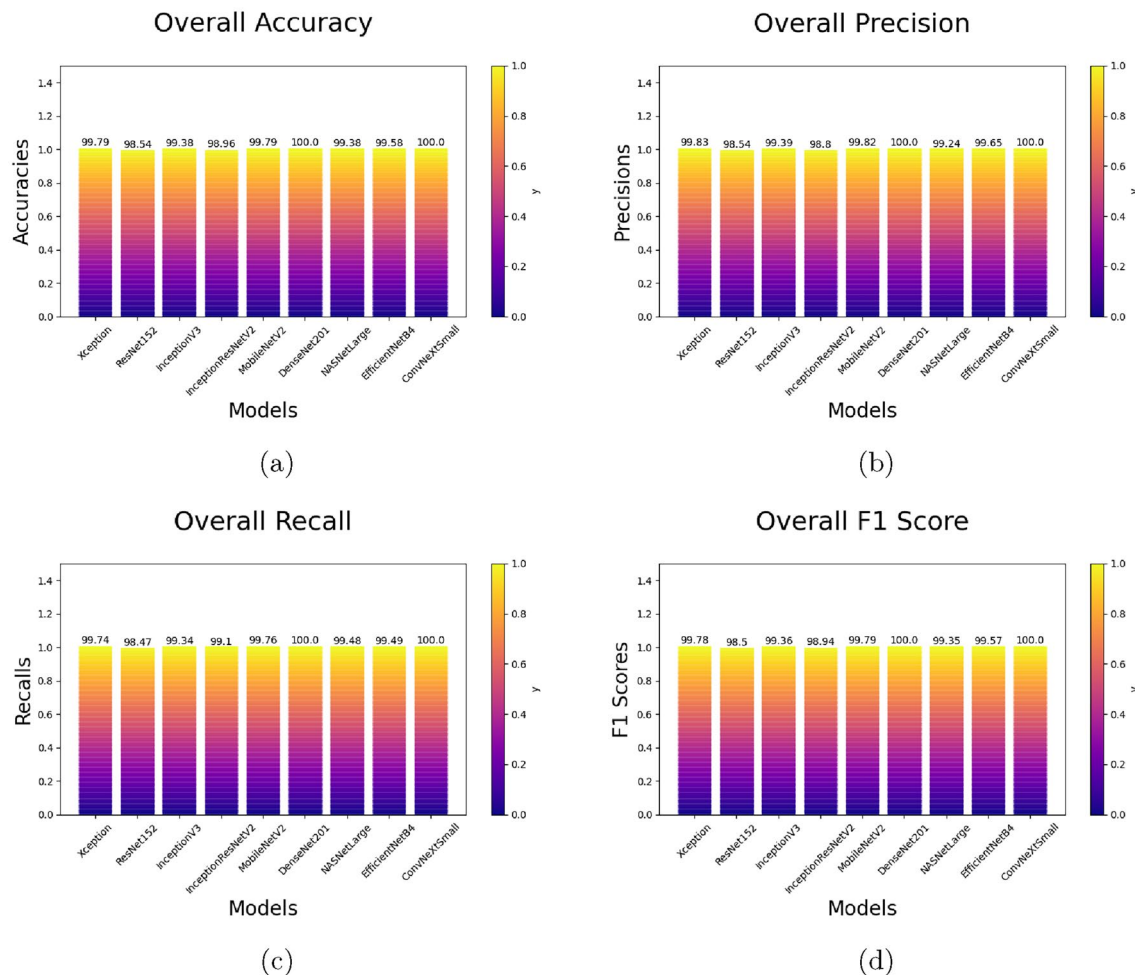


Fig. 22 Results of the Models depicting their overall accuracies, precisions, recalls and f1 scores on IBIB PAN Dataset: **a** Overall Accuracies of the models, **b** Overall Precisions of the models, **c** Overall Recalls of the models, **d** Overall F1 Scores of the models

used for all the models. This helps to achieve faster convergence, adapt to the learning rates for individual parameters, and handle sparse gradients.

The results given by these models along with some additional information are shown in Table 5

These models gives us very high accuracy scores respectively. The results show that these models are very reliable for this job. In the above listed models in Table 5, we can observe that DenseNet201 and ConvNeXtSmall both provides the best results. Whereas ResNet152 and InceptionResNetV2 provides the worst results. Table 5 also highlights the parameters of these models in our research work.

Table 6 provides a detailed results of these models. It also provides a detailed classwise results of each model.

From the Table 6 we can observe that DenseNet201 and ConvNeXtSmall are the best performing model. The rest of the performance are almost giving nearly similar good results. All these high classification accuracies indicate that the models are highly reliable Fig. 22 provides a pictorial representation of the results as depicted in Table 6.

For the train and validation datasets, the Loss and Accuracy Curves for each model are shown in Figs. 23, 24, 25, respectively. They also show the models' ROC-AUC curves and confusion matrix.

The models' loss curves show that there isn't a severe overfitting problem in any of the three models. As seen by Table 6, the accuracy curves represent the accuracy scores. Although some classifications are wrong, the Confusion Matrices demonstrate that the proportion of wrong to correct classifications is quite small, indicating that these models are reliable for the given job. The final ROC curve that each model produced on the test dataset demonstrates how low the False Positive rate is. The models' remarkably effective ability to classify test images into one of the two classes is indicated by the ROC-AUC Curves in Figs. 23, 24, 25 shifting to the upper left corner. This suggests that the preprocessing approach that has been suggested is highly effective for the given problem.

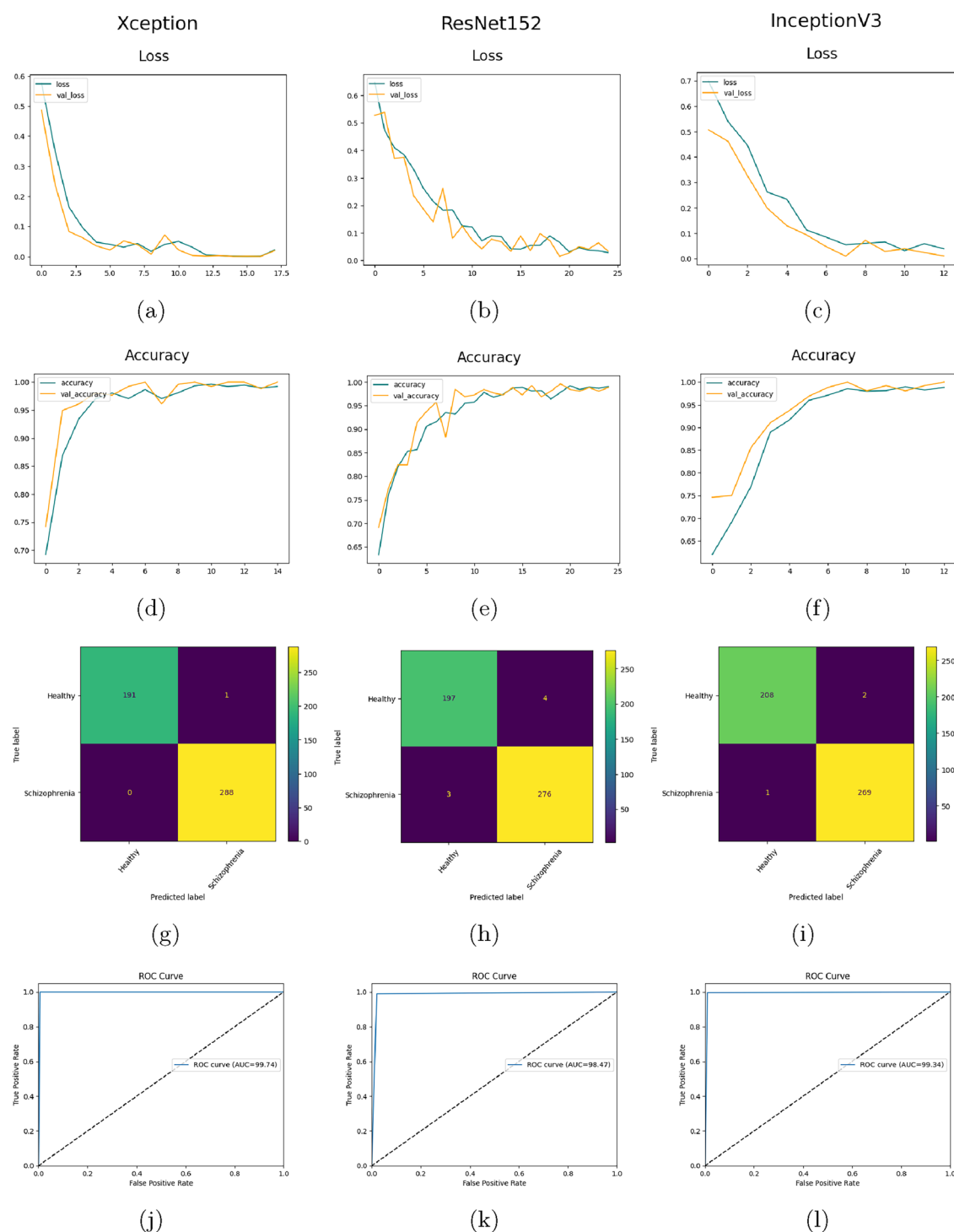


Fig. 23 Results of the Models Xception, ResNet152 and InceptionV3 on IBIB PAN Dataset: **a** Loss Curve of Xception, **b** Loss Curve of ResNet152, **c** Loss Curve of InceptionV3, **d** Accuracy Curve of Xception, **e** Accuracy Curve of ResNet152, **f** Accuracy Curve of InceptionV3, **g** Confusion Matrix of Xception, **h** Confusion Matrix of ResNet152, **i** Confusion Matrix of InceptionV3, **j** ROC-AUC Curve of Xception, **k** ROC-AUC Curve of ResNet152, **l** ROC-AUC Curve of InceptionV3

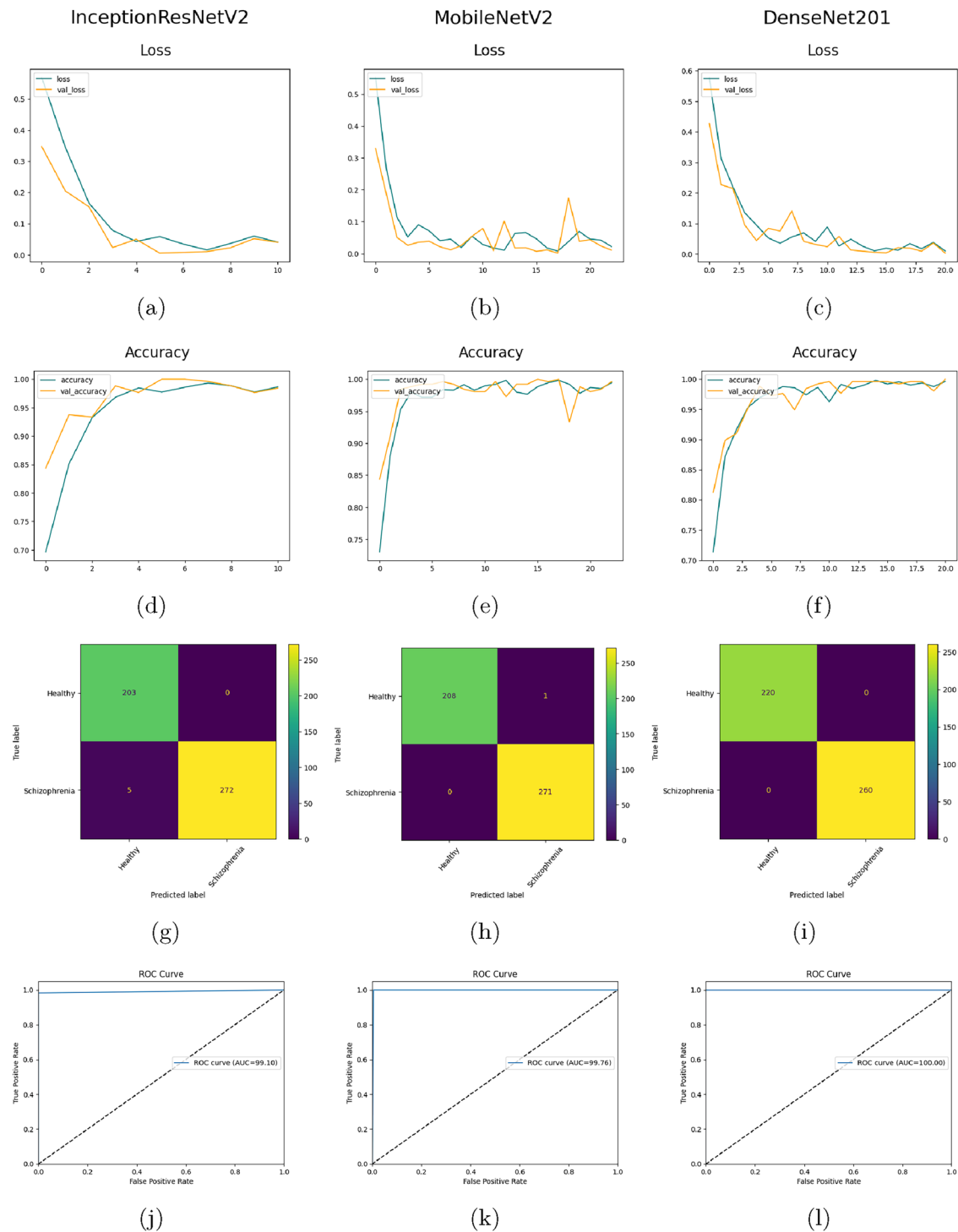


Fig. 24 Results of the Models InceptionResNetV2, MobileNetV2 and DenseNet201 on IBIB PAN Dataset: **a** Loss Curve of InceptionResNetV2, **b** Loss Curve of MobileNetV2, **c** Loss Curve of DenseNet201, **d** Accuracy Curve of InceptionResNetV2, **e** Accuracy Curve of MobileNetV2, **f** Accuracy Curve of DenseNet201, **g** Confusion Matrix of InceptionResNetV2, **h** Confusion Matrix of MobileNetV2, **i** Confusion Matrix of DenseNet201, **j** ROC-AUC Curve of InceptionResNetV2, **k** ROC-AUC Curve of MobileNetV2, **l** ROC-AUC Curve of DenseNet201

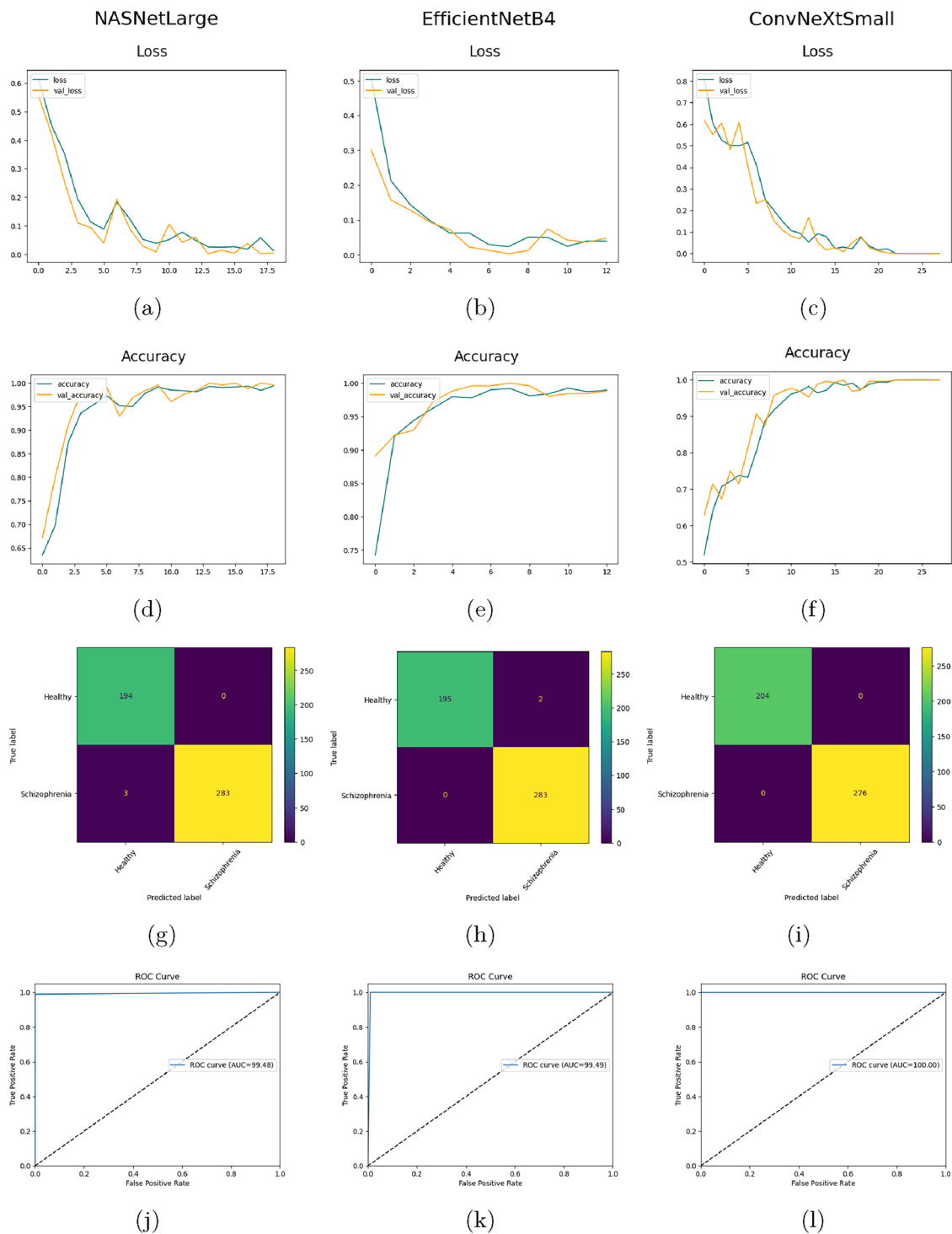


Fig. 25 Results of the Models NASNetLarge, EfficientNetB4 and ConvNeXtSmall on IBIB PAN Dataset: **a** Loss Curve of NASNetLarge, **b** Loss Curve of EfficientNetB4, **c** Loss Curve of ConvNeXtSmall, **d** Accuracy Curve of NASNetLarge, **e** Accuracy Curve of EfficientNetB4, **f** Accuracy Curve of ConvNeXtSmall, **g** Confusion Matrix of NASNetLarge, **h** Confusion Matrix of EfficientNetB4, **i** Confusion Matrix of ConvNeXtSmall, **j** ROC-AUC Curve of NASNetLarge, **k** ROC-AUC Curve of EfficientNetB4, **l** ROC-AUC Curve of ConvNeXtSmall

Table 7 Performance comparison of the proposed methodology with state-of-the-art methods. Results are in %

Method	Approach	Acc
Sun et al. [12]	Fuzzy Ensemble, FFT	99.22, 96.34
de Miras et al. [64]	SVM, RF, LR, kNN and DT	89, 87, 86, 86, 68
Shoeibi et al. [10]	CNN-LSTM	99.25
Soria et al. [11]	eXtreme Gradient Boosting	94.62
Bae et al. [23]	RF method	96.00
Lee et al. [21]	SVM	85.00
Proposed Methodology	Preprocessing with Fused Power Spectrograms and Xception	99.79
	ResNet152	98.54
	InceptionV3	99.38
	InceptionResNetV2	98.96
	MobileNetV2	99.79
	DenseNet201	100.00
	NASNetLarge	99.38
	EfficientNetB4	99.58
	ConvNeXtSmall	100.00

Bold values indicate the results obtained by us on the same dataset

5.4 Comparison with state of the art methods

This section compares the suggested approach with the approaches from the literature review. It is determined that, when employing the suggested strategy, the majority of the models have accuracy scores that are close to or greater than others. Table 7 presents an example of a comparison between our proposed model and others.

In our experiments using the proposed approach, DenseNet201 and ConvNeXtSmall achieves the best performance accuracy of 100.00% as compared to the other methods or individual classifiers. Methods proposed by [12] and [10] are based on the deep CNN methods and achieve accuracy of (99.22% and 96.34%), and 99.25%, respectively. [64] have attained 89%, 87%, 86%, 86% and 68% accuracy by using SVM, RF, LR, kNN and DT for the classification of brain EEG signals.

5.5 Data visualization

In this section, we take help from two data visualization tools to visually show some results of the proposed method they are GradCAM t-SNE plots.

5.5.1 GradCAM analysis

In this work, we have applied GradCAM, a method that, according to [65], generates a gradient-weighted class activation map, to generate visual representations of the model predictions. These illustrations aid in illuminating the decision-making process of neural networks. Using the models employed in this work, we have used GradCAM to create visualizations for the fused spectrogram pictures for the IBIB PAN dataset, as Fig. 26 illustrates. It is clear that the different models focus on different regions of the pictures, which implies that different learners take in different and complimentary information according to their own distinctive architectures.

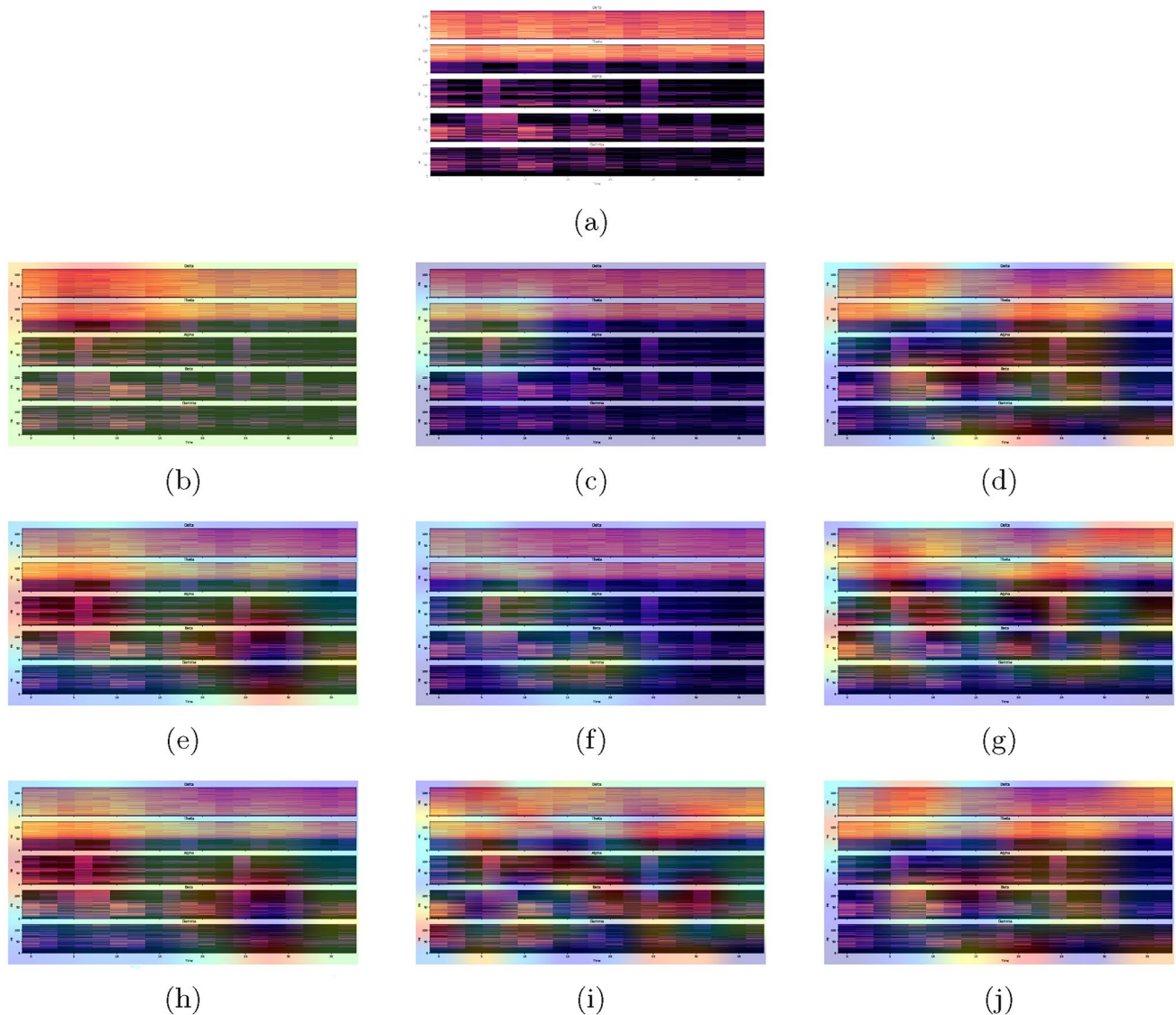


Fig. 26 GradCAM visualization of the fused spectrogram images on the IBIB PAN dataset for the models, where **a** Original Image **b** Xception GradCAM, **c** ResNet152 GradCAM, **d** InceptionV3 GradCAM, **e** InceptionResNetV2 GradCAM, **f** MobileNetV2 GradCAM, **g** DenseNet201 GradCAM, **h** NASNetLarge GradCAM, **i** EfficientNetB4 GradCAM, **j** ConvNeXtSmall GradCAM

5.5.2 t-SNE plots

According to [66], t-SNE (t-Distributed Stochastic Neighbor Embedding) is a prominent dimensionality reduction approach that can be used to show high-dimensional data in a lower-dimensional space. The program initially converts the high-dimensional Euclidean distances between the data points into similarity-indicating conditional probability scores. SNE (Stochastic Neighbor Embedding) on the data points is used to achieve this. The Eq. 7 is used to define the conditional probability $P_{j|i}$, which indicates how similar data points x_j and x_i are to one another.

$$P_{ij} = \frac{\exp(-\frac{\|x_i - x_j\|^2}{2\sigma_i^2})}{\sum_{k \neq i} \exp(-\frac{\|x_i - x_k\|^2}{2\sigma_i^2})} \quad (7)$$

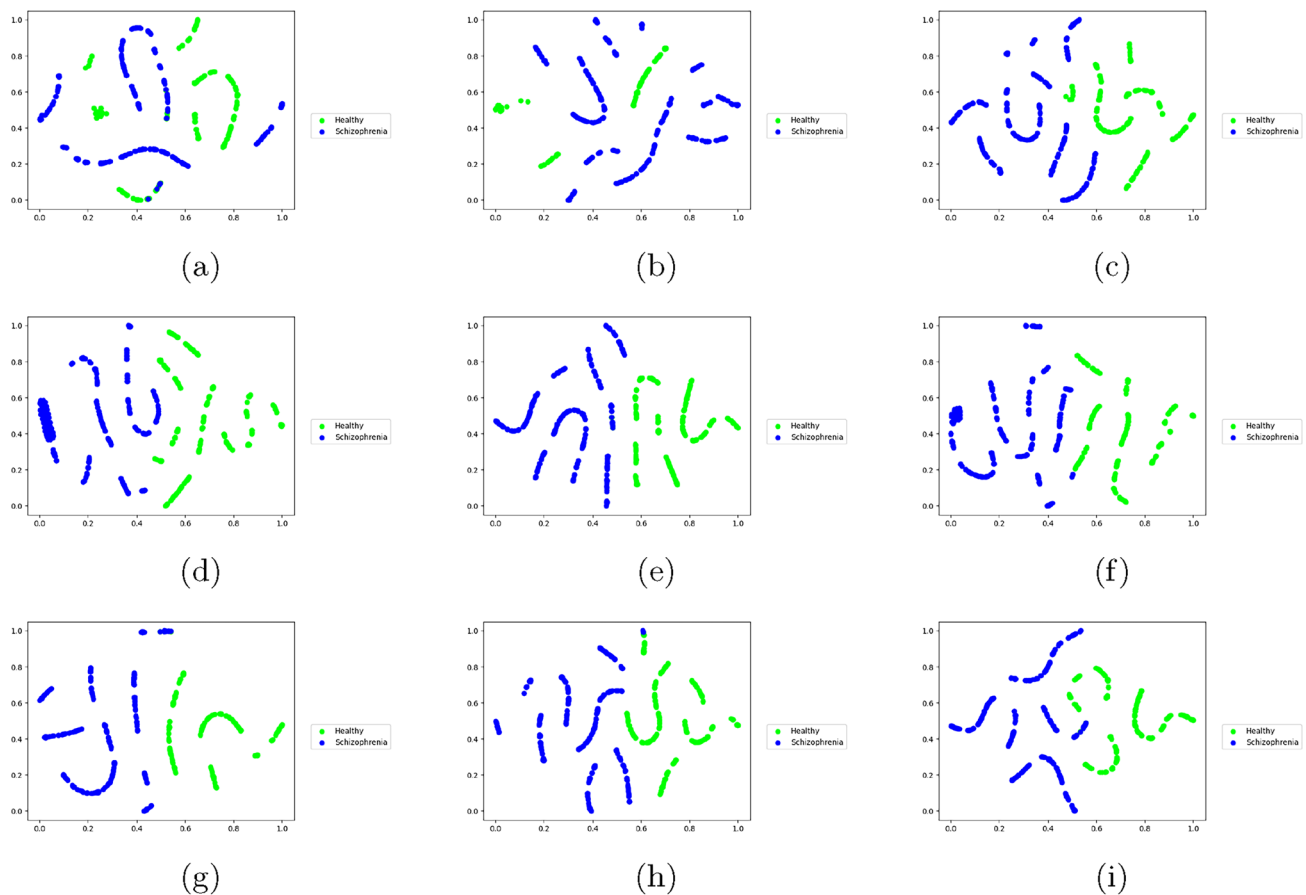


Fig. 27 t-SNE visualization for the models on the IBIB PAN dataset, where **a** Xception t-SNE, **b** ResNet152 t-SNE, **c** InceptionV3 t-SNE, **d** InceptionResNetV2 t-SNE, **e** MobileNetV2 t-SNE, **f** DenseNet201 t-SNE, **g** NASNetLarge t-SNE, **h** EfficientNetB4 t-SNE, **i** ConvNeXtSmall t-SNE

We observe that the images corresponding to the different emotions are clearly separated into different clusters of points. The t-SNE plot visualizations for the images from the IBIB PAN dataset by the classifiers are presented Fig. 27.

6 Conclusion and future scope

This study underscores the pivotal role of accurate neurophysiological monitoring in diagnosing and managing brain disorders, focusing particularly on Schizophrenia. The challenges in precise identification, especially within intricate neural structures, highlight the necessity for advanced methodologies. The novel pre-processing technique introduced here for feature extraction and image generation has shown promising results in enhancing the diagnostic potential of CNN-based architectures.

Our comprehensive exploration of various CNN models-DenseNet201, ResNet152, MobileNetV2, NasNetLarge, EfficientNetB4, ConvNeXtSmall, Xception, InceptionV3, and InceptionResNetV2 - has illuminated their varied benefits in classifying EEG data from the IBIB PAN dataset. These models, known for their efficacy in diverse classification tasks, offer unique advantages in addressing specific complexities within brain disorders.

The findings of this research hold significant promise for medical practitioners and researchers, providing valuable insights for improved diagnosis and treatment planning in the realm of brain disorders. By leveraging the outcomes of this study, professionals can enhance their understanding and approach toward these conditions, potentially leading to more effective interventions and personalized care.

Looking ahead, the future of neurophysiological monitoring and classification can be enriched further by exploring more complex and robust approaches like ensemble approaches, fusion techniques, and collaborative models. Implementing these advanced strategies can potentially amplify the accuracy and reliability of diagnostic tools, ultimately fostering more nuanced and tailored interventions for individuals affected by brain disorders.

This study contributes a significant step forward in leveraging CNN-based models and innovative preprocessing techniques, offering a pathway toward more robust diagnostic methodologies and treatment strategies for brain disorders in clinical practice.

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Data availability Have been made publicly available through the above GitHub link.

Materials availability Publicly Available.

Code availability Will be made publicly available through a GitHub link.

Declarations

Ethics approval and consent to participate Authors declare there is no Ethical problems and give their full consent for participation.

Consent for publication Authors give their full consent for publication.

Competing interests The authors declare no competing interests.

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